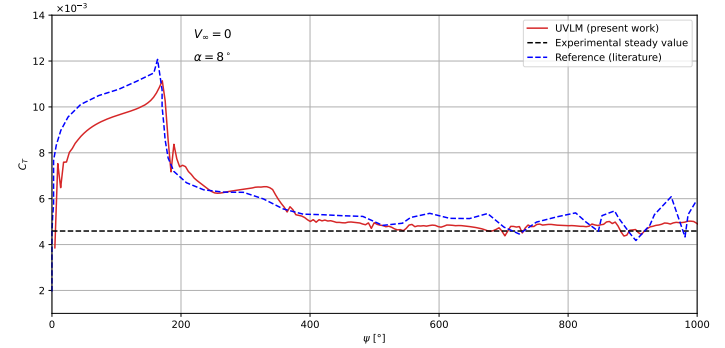
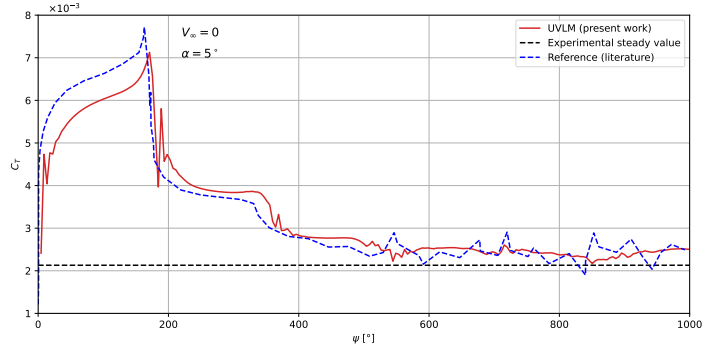


Thrust coefficient vs. blade rotation angle for a two-blade rotor in hover ($\alpha = 5^\circ$, $\alpha = 8^\circ$)



The figures show the evolution of the thrust coefficient C_T as a function of the blade rotation angle ψ for angles of attack of 5° and 8° . The overall trend closely matches the reference solution, with the only notable difference being a slight underestimation of C_T during the rising phase preceding the peak.

After half a rotor revolution, each blade approaches the starting vortex shed by the other blade. As soon as the two wakes begin to overlap, the C_T drops abruptly. This interaction generates complex local flow conditions that, over time, lead to numerical instability. When wake vortex segments come very close to each other, the singular behavior inherent to the Biot–Savart law produces high induced velocities, triggering the classical numerical instabilities typical of singularity-based methods. This explains the noise observed in the C_T evolution.

Nevertheless, the mean steady-state value predicted by the model agrees well with the experimental measurements for both angles of attack.